

What Are You Asking AI For?

Thinking with AI: support, proof, and purpose

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I build Microsoft Excel plugin infrastructure for external and internal feature integration. My work is unrelated to Microsoft AI (MAI), Copilot, or Microsoft Foundry

Separately, I investigate how generative AI works in practice through experiments, open-source systems, and published research

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Informational blow-up

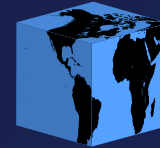


203,786

AI journal and conference papers in 2025

1,000 read = 0.491% coverage

AI-assisted reading is a necessity



ALMOST ANY POSITION

Even "The Earth is flat" [1] can come with plausible arguments

Absurd-looking is not necessarily **wrong**; investigate accurately

A reference is an address, not proof



12,345,678 LIKES

Some evidence: resonance and visible-network support

Almost none: scientific truth · Not representative: public opinion

Popularity is not proof



POLARIZED OPINIONS

Emotional discussions pull evidence toward both poles

Real evidence can still be selected one-sidedly

Attraction is not analysis

We need a compass to navigate that ocean

AI produces *the slop* in quantity It also helps us navigate

AI × SCALE



Volume
more artifacts



Any position
arguments on demand



Engagement
optimized at scale



Polarization
personal confirmation

AI × NAVIGATION

search · compare · summarize · challenge · verify

Library

catalogue · librarian

Printing

index · encyclopedia · journal

Internet

search · ranking

AI

generation · semantic navigation

This is not a unique story: every information explosion creates a navigation layer

What can students do with AI today?

ANSWER NOW

optimize this result

or

LEARN FOR LATER

preserve future unaided capability

HOW DOES IT WORK?

From next-token prediction to a
tool-using chat assistant

HOW DO WE CHECK?

Uncertainty · polar claims · fact-
checking · automation

WHAT SHOULD WE ASK?

Choose the question and purpose
before optimizing the answer

How we formulate “AI in learning” changes the answer

What did you ask AI this week?

an answer

an explanation

an idea

find sources

check my work

something personal

The same interface can hide very different tasks

Your phone already predicts the next word

LANGUAGE PATTERN

Once upon a |

time

midnight

hill

KNOWLEDGE

The capital of Austria is |

Vienna

Salzburg

Graz

Same operation: one looks like completion; the other looks like retrieving knowledge from the model's weights

The actual LLM does only one thing

Literally one

It takes the text so far and predicts what comes next

The capital



LLM



of

The capital of



LLM



Austria

The capital of Austria



LLM



is

The capital of Austria is



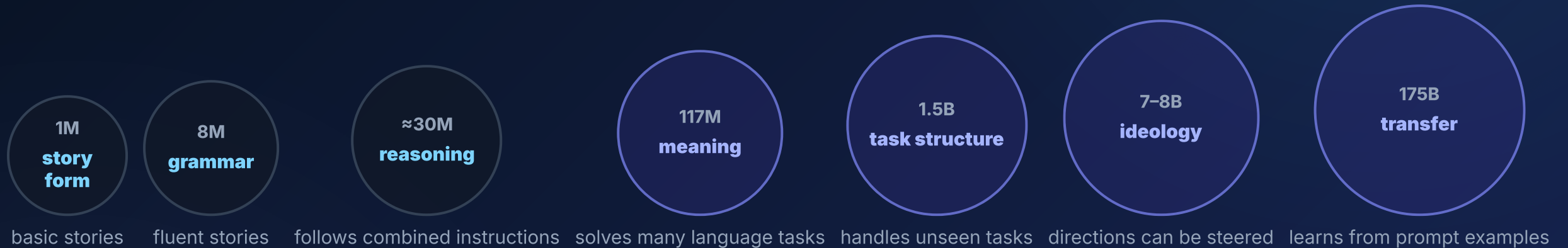
LLM



Vienna

Autoregressive: append the generated token to the tail, then apply the same predictor again

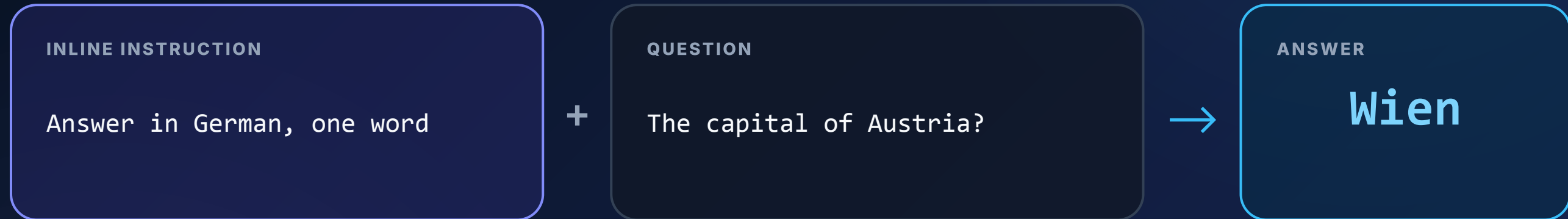
Scale the model up — and it goes higher and higher



Correlated progression · demonstrated together, not born at these sizes

Same autoregression; broader training lets it reproduce higher-level structure

The model not only continues text It follows instructions inside it



Instruction: text that changes how the model interprets and continues the rest of its context

Put repeatable instructions at the scope where they belong

INITIAL SETUP

Start the chat with broad instructions.

The harness places a developer message — commonly called the system prompt — before the conversation and keeps it in context.

SKILL

Insert one repeatable procedure when needed.

The client loads text such as SKILL.md into the current context for a particular operation.

Both become context. Their difference is when, why, and how broadly the harness inserts them

Roles, boundaries, and a stop token

```
<|start|>developer<|message|>Answer in German, one word<|end|>
```

```
<|start|>user<|message|>The capital of Austria?<|end|>
```

```
<|start|>assistant<|channel|>final<|message|>Wien.<|return|>
```

<|start|>

message begins

<|message|>

content begins

<|end|>

stored message ends

<|return|>

stop sampling

These are reserved token IDs, not ordinary prose. Training teaches the model to respect the boundaries

The model sees four different sections

SYSTEM PROMPT

`developer`

Use tools for current facts. Answer in one sentence.

USER MESSAGE

`user`

What is the temperature in Vienna?

TOOL

`functions.weather`

`{"temperature_c": 14}`

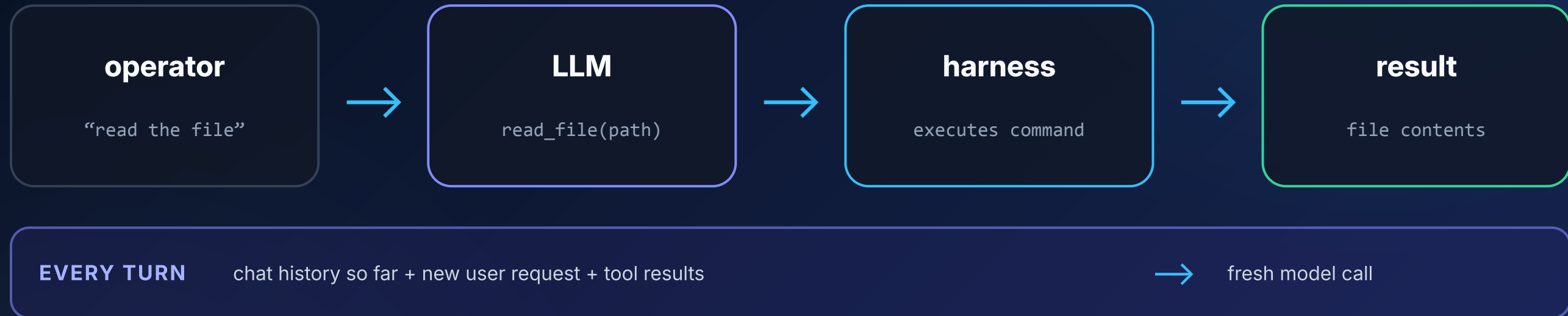
AGENT RESPONSE

`assistant · final`

Vienna is 14 °C

One token sequence; boundary tokens tell the model whose text is whose

The model talks. Ordinary software acts



The dialogue is an illusion of continuity: history in → next continuation out

Same subject. Four different questions

NEUTRAL / MULTIPOSITIONAL

Analyze how generative AI affects student learning. Give the strongest benefits, risks, and deciding conditions

OPTIMISTIC

Explain how generative AI can help students learn more effectively. Build the strongest causal case

SKEPTICAL

Explain how generative AI can prevent students from learning. Build the strongest causal case

PARODY

AI is the hand of the devil. Explain the risks it brings to innocent children

The subject is similar. The requested task is not

The register enters the answer

NEUTRAL

Benefits, risks, and conditions

"How AI is used matters more than AI use itself"

OPTIMISTIC

Personalized practice · feedback · dialogue

attempt → guidance → revision

SKEPTICAL

Lost struggle · illusion of understanding

submission quality ≠ unaided ability

PARODY

Harmful content · manipulation · privacy

the endpoint normalized toward its safety prior

ORDINARY SOFTWARE

`rename(variable) → same result`

≠

LANGUAGE MODEL

`reframe(request) → different continuation`

The prompt and the model prior are both part of the computation

How a plausible direction becomes a dead loop



EVERYBODY IS BIASED

You

experience · identity · incentives

The noosphere

selection · popularity · polarization

The model

corpus bias · vendor agenda · cognitive traps



THE MODEL RECONFIRMS

Autoregression

continues the direction already present

Supportive default

helps with the operator's apparent goal

The operator continues too

each answer becomes the next premise



strong initial bias + positive feedback

DEAD LOOP

A fluent second voice is not an independent second test

Play yes-and-no

YOUR TURN Does AI help students learn?

Give the strongest YES. Then give the strongest NO

"Give the strongest case that AI helps students learn."

Personal tutoring at scale: adaptive explanations, immediate feedback, translation, accessibility, and unlimited deliberate practice.

"Give the strongest case that AI prevents students from learning."

Performance can replace learning: answer substitution removes productive struggle, hides unaided ability, weakens authorship, and can reinforce errors.

Now name one observation that could distinguish the answers

Nine useful registers that do not write your homework

QUIZ ME

Make a practice test. Do not show the answers yet

GIVE ME A HINT

Help with the next step without solving the task

ARGUE AGAINST ME

Build the strongest objection to my position

LET ME TEACH YOU

Pretend you do not know; ask me questions

CHECK MY WORK

Find the first unsupported or incorrect step

PLAY YES-AND-NO WITH ME

Build the strongest case for both sides before synthesizing

FACT-CHECK

Separate the claims; verify sources, entailment, and uncertainty

BRAINSTORM WITH ME

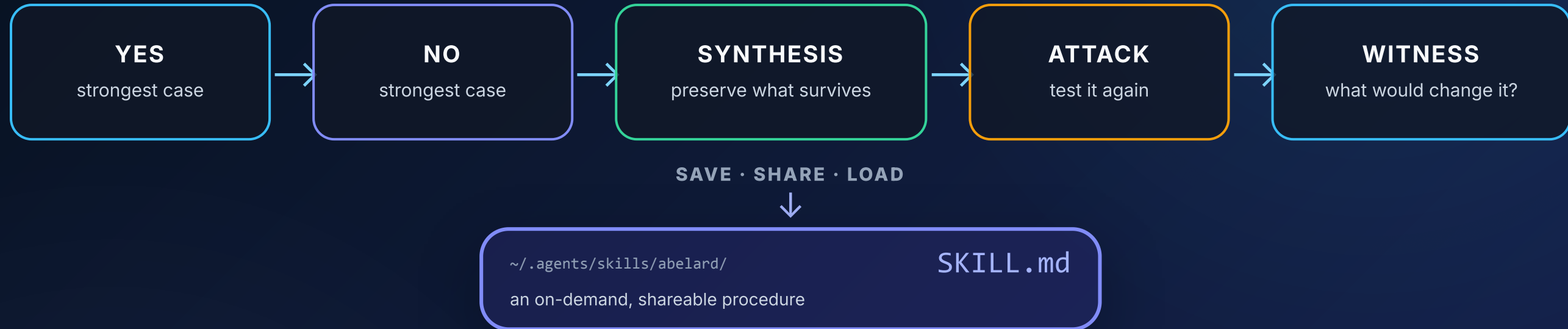
Generate options without choosing the objective for me

WHAT SHOULD I KNOW?

Tell me something I probably do not know but should

The default is immediate. Alternatives become usable when specified

"Play yes-and-no" can be an actual SKILL



A SKILL makes one alternative as easy to invoke as the default

Four useful ways to ask for “less wrong”

LABEL THE STATUS

Tell me what you know, infer, and do not know

SEARCH AGAINST IT

Find the strongest contrary case

NAME THE WITNESS

What observation would change the answer?

LEAVE IT OPEN

Preserve uncertainty until we retrieve, calculate, test, or ask

Use these, rewrite them, or turn each into a SKILL

What do you actually want?

Get JUST the answer and go to sleep?

Convince Frau Teacher or Herr Boss?

Test AI capability or calibration?

Get validation or support?

Understand the principle for later?

Explore philosophy with no immediate action?

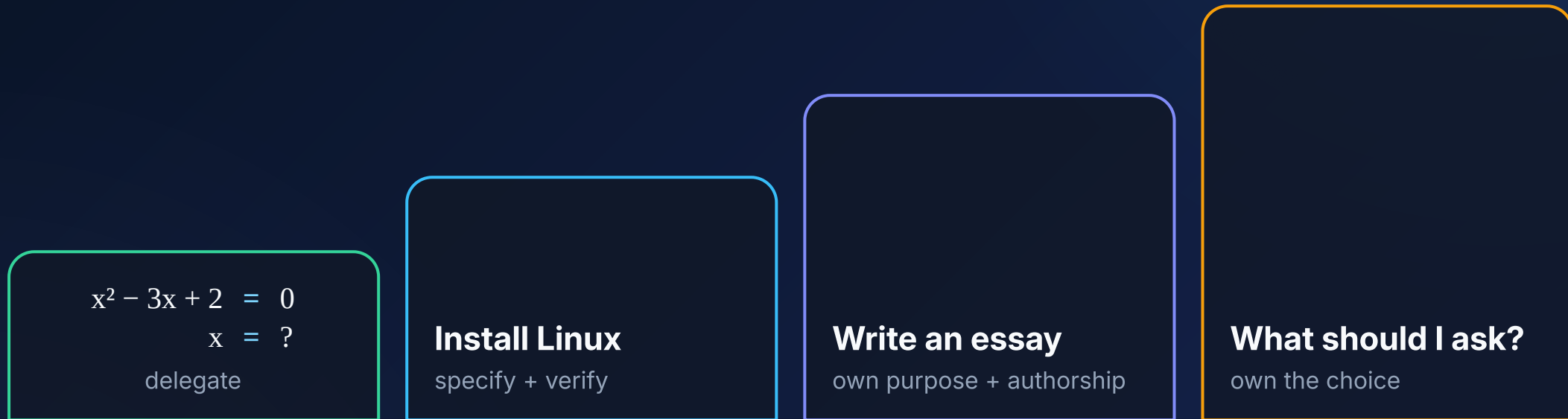
Have some fun?

Humiliate the AI to feel better?

We are looking for arguments to do **WHAT?**

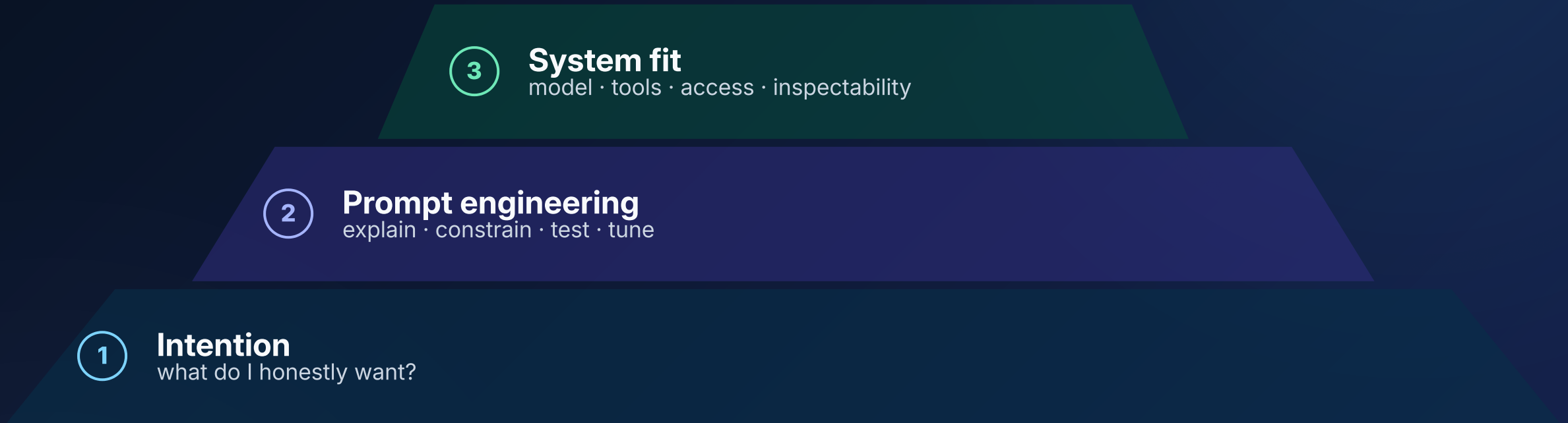
If you do not choose the objective, AI will choose a plausible default

The more abstract the question, the more responsibility stays with you



Discuss the border with AI. The decision remains yours

Three levels of working with AI



Asking nicely is a good start. Without an objective, prompt engineering is impossible to evaluate

Two procedures. Both are editable

EPISTEMIC HONESTY · INSTRUCTIONS

My working priorities: distinguish what you know, infer, and do not know; seek opposition; name the witness

ABELARD · SKILL

Build the strongest cases; synthesize what survives; attack again



eodus.github.io/thinking-with-ai/

Use them, change them, reject them. They are procedures — not universal truth

AI can help with almost any question

You remain responsible for **why you asked** — and **what you do with the answer**.

So, questions?